



PROJECT REPORT

Predicting Electric Vehicle Charging Infrastructure in Southern Ontario Using Machine Learning

1. Introduction

The global shift toward sustainable transportation has significantly increased the adoption of electric vehicles (EVs). Governments worldwide, including Canada, are promoting EV usage to reduce carbon emissions and dependency on fossil fuels. Southern Ontario, being one of the most populated and economically active regions in Canada, has witnessed a rapid increase in EV adoption.

However, the development of EV charging infrastructure has not kept pace with this growth. Insufficient and uneven distribution of charging stations can lead to range anxiety among EV users and hinder further adoption.

This project focuses on predicting the demand for EV charging infrastructure in Southern Ontario using machine learning techniques. By analyzing factors such as population density, EV registrations, and existing charging stations, the project aims to identify areas requiring infrastructure expansion.

2. Problem Statement

With the rapid increase in EV adoption, the current charging infrastructure is insufficient and unevenly distributed across Southern Ontario. This creates challenges for EV users and policymakers.

The main problem addressed in this project is:

How can we predict future demand for EV charging stations to support efficient infrastructure planning?

3. Objectives

The objectives of this project are:

- To analyze EV growth trends in Southern Ontario
 - To identify key factors influencing charging infrastructure demand
 - To build machine learning models for demand prediction
 - To highlight high-demand regions for future infrastructure development
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4. Literature Review

Several studies have explored EV adoption and infrastructure planning. Research indicates that factors such as population density, income levels, and urbanization significantly influence EV usage.

Machine learning techniques, including Linear Regression and Random Forest, have been widely used for predictive modeling due to their ability to handle complex datasets.

Previous studies have also used Geographic Information Systems (GIS) for mapping charging station distribution. However, limited work has been done specifically for regional prediction in Southern Ontario, which this project aims to address.

5. Methodology

5.1 Data Collection

Data for this project is collected from publicly available sources, including:

- EV registration statistics
 - Population density data
 - Charging station locations
 - Urban and suburban classifications
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5.2 Data Preprocessing

The collected data undergoes preprocessing steps such as:

- Handling missing values
- Normalization of numerical data

- Encoding categorical variables
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5.3 Feature Selection

The following features are used in the model:

- Population density
 - Number of EVs
 - Distance to nearest charging station
 - Traffic density (if available)
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5.4 Model Development

Two machine learning models are implemented:

1. **Linear Regression**
 - Used as a baseline model
 - Assumes a linear relationship between variables
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5.5 Model Evaluation

The performance of the models is evaluated using:

- Mean Absolute Error (MAE)
 - Root Mean Square Error (RMSE)
 - R-squared (R^2) score
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6. Results

Key findings include:

- High demand in urban areas such as Toronto, Mississauga, and Hamilton
- Increasing demand in suburban regions
- Uneven distribution of existing infrastructure

Visualizations such as bar charts and scatter plots highlight the relationship between population and infrastructure demand.

7. Discussion

The results demonstrate that EV infrastructure planning must be proactive and data-driven. Urban centers require immediate expansion, while suburban areas need strategic planning to meet future demand.

The model successfully identifies high-demand regions but has limitations due to data constraints.

8. Challenges and Limitations

- Limited availability of real-time data
 - Lack of detailed traffic data
 - Assumptions in EV growth projections
 - Static dataset without time-based analysis
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9. Future Work

Future improvements to this project could include:

- Integration of real-time traffic and usage data
 - Implementation of time-series forecasting models
 - Use of advanced algorithms such as XGBoost
 - Development of interactive dashboards
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10. Conclusion

This project presents a machine learning-based approach to predict EV charging infrastructure demand in Southern Ontario. The findings highlight the importance of data-driven decision-making in infrastructure planning.

The proposed model provides valuable insights for policymakers and urban planners, helping to support the transition toward sustainable transportation.

11. References

1. Government of Canada – EV Statistics
2. Ontario Open Data Portal
3. Research papers on EV infrastructure and machine learning
4. IEEE publications